

QUESTION 1:

Design a class to represent a bank account. Include the following members.

Data Members

- Name of the depositor.
 - Account number.
 - Type of account.
 - Balance amount in the account
- Methods.
- To assign initial values.
 - To deposit an amount.
 - To withdraw an amount after checking balance.
 - To display the name and balance
- Incorporate a constructor to provide initial values.

CODE:

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

class BankAccount {
private:
    string depositorName;
    long accountNumber;
    string accountType;
    double balance;

public:
    BankAccount(string name, long accNum, string type, double initialBalance) {
        depositorName = name;
        accountNumber = accNum;
        accountType = type;
        balance = initialBalance;
    }

    void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            cout << "deposited amount: $" << amount << endl;
        } else {
            cout << "Error: Deposit amount must be positive." << endl;
        }
    }
};
```

```

    }
}

void withdraw(double amount) {
    if (amount > balance) {
        cout << "Error: Insufficient balance." << endl;
    } else if (amount <= 0) {
        cout << "Error: Withdrawal amount should be positive." << endl;
    } else {
        balance -= amount;
        cout << "withdrawn amount: $" << amount << endl;
    }
}

void display() {
    cout << "Name: " << depositorName << endl;
    cout << "Current Balance: $" << fixed << setprecision(2) << balance <<
endl;
}
};

int main() {
    BankAccount myAcc("Riddhima Gupta", 987654321, "Savings", 1000.00);

    myAcc.display();

    myAcc.deposit(500.50);
    myAcc.withdraw(200.00);
    myAcc.withdraw(2000.00);

    myAcc.display();
}

```

OUTPUT:

```

Name: Riddhima Gupta
Current Balance: $1000.00
deposited amount: $500.50
withdrawn amount: $200.00
Error: Insufficient balance.
Name: Riddhima Gupta
Current Balance: $1300.50

```